

ARTHUR REALISAN JR., CE

PRC Registered Civil Engineer | Structural & Cost Estimation Specialist

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90.35% CELE

BOARD RATING

Top 1% Nationwide (Out of 17,062)

EXECUTIVE SUMMARY

High-performing PRC Registered Civil Engineer with a Major in Structural Engineering and an elite 90.35% Civil Engineering Licensure Examination (CELE) rating (placing in the top 1% nationwide out of 17,062 examinees). Combines advanced structural finite element modeling expertise (MIDAS Gen, STAAD Pro, Autodesk CFD) with practical cost engineering, digital quantity takeoffs (Bluebeam Revu, Planswift), and rigorous budget auditing capabilities. Proven track record in national civil/structural engineering competitions and capstone research on aerodynamic wind efficiency for high-rise structures.

LICENSURE & ACADEMIC DISTINCTION

PRC Registered Civil Engineer

September 2026 CELE

- Board Examination Rating: 90.35% Overall** — Ranked in the upper percentile nationwide out of 17,062 examinees, performing within 1.6% of the national Top 10 threshold.

EDUCATION & CAPSTONE RESEARCH

Bachelor of Science in Civil Engineering (Major in Structural Engineering)

Sep 2022 – Jul 2026

New Era University | Quezon City, Philippines

Featured Research & Thesis Project

“The Structural Efficiency of Aerodynamic Building Forms for Coastal High-Rise Condominiums: A Case Study in Zambales”

- Multi-Software Workflow:** Integrated 3D Computational Fluid Dynamics (Autodesk CFD / Fusion 360) wind simulations with full 3D Finite Element Structural Modeling in MIDAS Gen to analyze a 75-meter (25-story) high-rise structure under extreme 200 km/h typhoon wind loads (NSCP Wind Zone II).
- Parametric Structural Comparison:** Modeled and evaluated 4 structural building configurations (1 Prismatic baseline and 3 Tapered variations) across 3,544 structural members to determine Demand-to-Capacity Ratios (DCR), story drift, and wind pressure distributions.
- Statistical Validation & Cost Optimization:** Conducted non-parametric hypothesis testing using IBM SPSS (Kruskal-Wallis H test, $p = 0.002$) proving that optimal aerodynamic tapering significantly lowers member stress (reducing critical member DCR from 0.939 to 0.909). Demonstrated that structural shape tailoring reduces lateral demand, allowing for optimized member cross-sections and lower material quantities without compromising structural safety.

HONORS, AWARDS & NATIONAL COMPETITIONS

Champion — NEU Excellence Quiz Bee Oct 2025

Won 1st place in the university-wide engineering competition, demonstrating advanced technical problem-solving speed and mastery of core structural and civil engineering principles.

Rank 11 Nationwide — MIDAS Academic Festival Quiz Bee Oct 2025

Placed 11th nationwide in a specialized technical competition focusing on structural analysis, finite element modeling concepts, and advanced engineering mechanics.

National Competitor — ASEP 8th StructWhiz Challenge May 2025

Represented New Era University in the prestigious national structural engineering competition organized by the Association of Structural Engineers of the Philippines (ASEP), tackling complex structural design challenges.

National Competitor — National Civil Engineering Quiz (NCEQ) Oct 2025

Competed against top civil engineering delegates nationwide across structural design, mathematics, hydraulics, and construction management.

1st Runner-Up — NEU Excellence Quiz Bee Oct 2024

Secured 2nd place in the university-wide technical competition, showcasing consistent academic excellence.

PROFESSIONAL EXPERIENCE

Cost Estimator – Intern (240 Hours) Jan 2026 – Mar 2026

Local Water Utilities Administration (LWUA) | Quezon City, Philippines

- Assisted senior estimators and engineers in preparing comprehensive cost estimates and detailed quantity takeoffs for public water supply infrastructure and civil works.
- Reviewed structural, civil, and hydraulic engineering plans to calculate precise material, labor, and equipment requirements in compliance with government agency standards.
- Evaluated contractor budget proposals, audited market material rates, and streamlined cost assessment spreadsheets.
- Conducted demographic forecasting and population projections in Microsoft Excel for Calatagan, Batangas to estimate future municipal water demand.

TECHNICAL SKILLS & SOFTWARE MATRIX

Structural Analysis & Design	MIDAS Gen, STAAD Pro, AutoCAD, Structural Framing Interpretation, Finite Element Analysis (FEA), Load Combinations (NSCP 2015, ACI 318, AISC), Demand-Capacity Ratio (DCR) Optimization.
Aerodynamics & Simulation	Autodesk CFD, Autodesk Fusion 360, Wind Pressure Mapping, Story Drift & Lateral Load Analysis.
Cost Estimation & QS	Bluebeam Revu (Digital Takeoffs/Markups), Planswift, Primavera P6, Quantity Takeoffs (m ³ , m ² , linear meters), Bill of Quantities (BOQ) Build-ups, Site Photo Progress Auditing.